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Predicting the Efficacy of Antiarrhythmic Agents for Interrupting Persistent Atrial Fibrillation According to Spectral Analysis of the Fibrillation Waves on the Surface ECG

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Background: Spectral analysis of the fibrillation waves was performed in patients with persistent atrial fibrillation (PAF) to clarify the usefulness of this method of predicting the efficacy of antiarrhythmic agents.
Methods and Results: The 59 patients with PAF were randomly assigned to pilsicainide (150 mg/day) or bepridil (200 mg/day) therapy for 4 weeks under optimal anticoagulation therapy. When the first therapy did not interrupt PAF, the drugs were changed in a cross-over manner. The fibrillation cycle length (FCL) was calculated using spectral analysis of the fibrillation waves on the body-surface ECG every 2 weeks. Pilsicainide and bepridil were effective in 19 and 20 patients, respectively. The FCL at the basic state was longest in the pilsicainide-effective group, moderate in the bepridil-effective group and shortest in the failure group ($P<0.05$). The change in FCL with drug administration (ΔFCL) was larger in the effective than in the ineffective cases ($P<0.01$). Successful interruption of the atrial fibrillation (AF) with pilsicainide could be expected for patients with a FCL >148 ms (sensitivity = 0.917, specificity = 0.612, $P=0.007$) and ΔFCL >41 ms (sensitivity = 0.875, specificity = 0.833, $P=0.001$).
Conclusions: The FCL reflects the electrophysiological properties of the AF substrate and is considered useful for predicting the efficacy of antiarrhythmic agents. (*Circ J* 2009; **73:** 1210–1218)

Key Words: Antiarrhythmic agent; Atrial fibrillation; Fibrillation wave; Spectral analysis

Although catheter ablation is a useful procedure for the management of atrial fibrillation (AF),[1,2] pharmacological interruption of AF is still important step in the management of persistent AF (PAF) in clinical practice. When seeking to interrupt AF, the drivers of the maintenance of fibrillation (ie, the multiple reentrant circuits along the atrial surface) should be interrupted by either conduction block or prolongation of the effective refractory period as a result of the administration of antiarrhythmic agents,[3] but the practical effect of interrupting AF would hardly be expected because the electrophysiological properties may vary among individual patients.[4,5] Further, there is not a feasible or objective index for the selection of antiarrhythmic agents. One report has focused on the changes in AF frequency as a result of the administration of drugs,[6] but the parameters were not available for the prediction of drug efficacy. In the present study, we hypothesized that spectral analysis of the fibrillation waves in the surface ECG would be an index for predicting the efficacy of antiarrhythmic agents for interrupting AF, because it can be considered to reflect the electrophysiological properties of the atrial arrhythmogenic substrate.[7,8] To test the hypothesis, the indices of spectral analysis of the fibrillation waves on the surface ECG were measured and compared among patients with PAF who were treated with either pilsicainide or bepridil.

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Methods

Study Population

The study population consisted of 59 consecutive patients with AF lasting longer than 2 weeks at the time of the enrollment. As shown in **Table 1**, there were 26 women and 33 men, and the mean age was 64±6 years. When sinus rhythm could not be achieved using any type of antiarrhythmic therapy, including direct current cardioversion, the patient was excluded from the evaluation because the AF was defined as chronic. Patients with obvious renal or liver dysfunction (creatinine >2 mg/dl or total bilirubin >2 mg/dl) were also excluded. Structural heart disease was diagnosed by routine echocardiography, and patients with obvious left ventricular dysfunction (left ventricular ejection fraction <40%) were excluded. All patients were prescribed an optimal dose of warfarin as anticoagulation therapy to prevent any cardiogenic embolisms. All studies were performed with the approval of the Clinical Studies and Ethics Committee of Kitasato University Hospital.

Pharmacological Cardioversion

In all patients, therapy to terminate AF was started after

Table 1. Clinical Characteristics of the Patients

	Total (n=59)	Pilsicainide-effective (n=19)	Bepridil-effective (n=20)	Failure (n=20)	P value
Age (years)	65±6	64±5	63±7	66±6	0.553
Gender (F/M)	26/33	9/10	7/13	10/10	0.596
LVEF (%)	63±6	63±5	64±6	62±6	0.777
LAd (mm)	42±6	41±7	43±5	41±6	0.656
FCL (ms)	150±17	166±17	147±12	138±10	<0.001*
FCL-CV (%)	17±5	21±6	17±4	14±3	<0.001*

P-values were calculated among 3 groups of pilisicainide-effective, bepridil-effective and failure patients.
*Statistical significance.
LVEF, left ventricular ejection fraction; LAd, left atrial dimension; FCL, fibrillation cycle length; FCL-CV, FCL coefficient of variation.

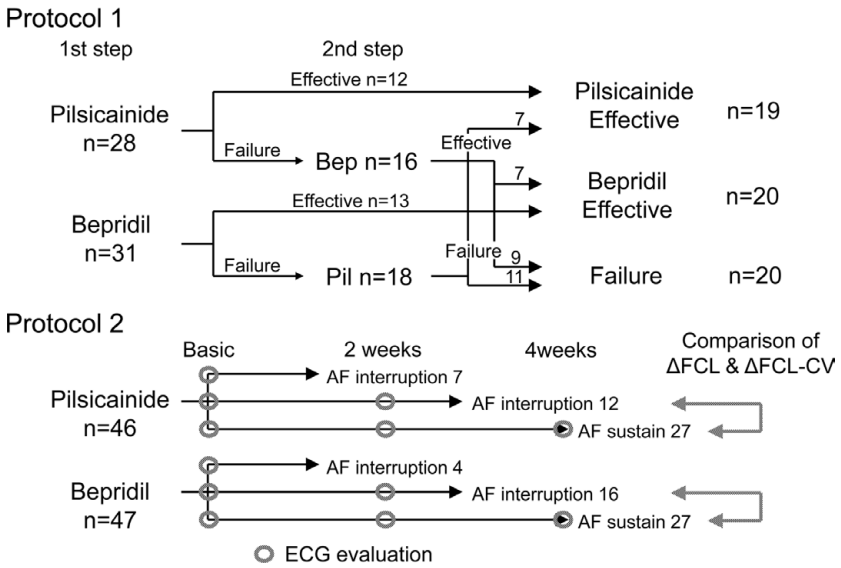

Figure 1. Results of pharmacological cardioversion and grouping. Of 59 patients, 19 were assigned to the pilsicainide-effective, 20 to the bepridil-effective, and the remaining 20 to the failure group (Protocol 1). AF-ECGs after the 2-week therapy could be evaluated in 12 effective and 27 failure patients with pilsicainide therapy, and in 16 effective and 27 failure patients with bepridil therapy, respectively. AF, atrial fibrillation; Bep, bepridil; Pil, pilsicainide; FCL-CV, fibrillation cycle length coefficient of variation.

optimal anticoagulation therapy or denial of any intra-atrial thrombi by transesophageal echocardiography. The patients were randomized to a pharmacological therapy to terminate AF: (1) 150 mg/day of oral pilsicainide for 4 weeks (representing class Ic antiarrhythmic agents)[9,10] or (2) 200 mg/day of oral bepridil for 4 weeks (representing class III antiarrhythmic agents).[11–13] These antiarrhythmic agents were selected because they are commonly used for AF treatment in clinical practice, at least in Japan, and interruption of AF was highly likely within a 4-week observation period.[9–13] If this first step did not achieve cardioversion to sinus rhythm, the drugs were changed in a cross-over manner after a washout period of the first drug for 2 weeks, and the patients underwent a further 4-week pharmacological treatment with other medication. If AF was interrupted by either of the antiarrhythmic agents, the patients were classified into the "pilsicainide-effective group" or "bepridil-effective group". If both medications failed to achieve sinus rhythm, other therapies, including a combination of antiarrhythmic agents or direct current cardioversion, were assigned to terminate the AF and those patients were classified into the "failure group".

ECG and Spectral Analysis

The body-surface ECG was recorded every 2 weeks until recovery of sinus rhythm, and spectral analysis of the fibrillation waves was performed. The ECG traces in lead V_1 were digitally stored in the computer system at a sampling rate of 1 kHz. The QRS-T complex template was subtracted using a template matching algorithm[7,8] to extract the pure fibrillation wave for the analysis. A frequency analysis was performed off-line on a microcomputer (Bimutas II, Kissei Co Ltd, Matsumoto, Japan) as previously described.[8,14] The frequency analysis of the subtracted ECG involved 3 steps: (1) band-pass filtering, (2) application of a Hamming window and (3) a 4096-point fast Fourier transformation. A 50% overlap of adjacent spectral analyses allowed the use of an average of 20 epochs of analyses within a single 44-s data set.[8] The power spectra were quantified by measuring the peak frequency signal with the maximum magnitude derived from each epoch. The peak frequency of the spectrum in the 3–12-Hz range was converted to a cycle length (CL in ms = 1,000/frequency), defined as the fibrillation CL (FCL), and was averaged from 20 epochs. The temporal variability was expressed as the FCL coefficient of variation (FCL-CV) to be used as an index for the variation in the FCL during the time-course of the recording.[8]

Comparison of the FCL Data

Protocol 1 (Comparison of the Data at Baseline) The study population was divided into 3 groups according to the results of the drug efficacy testing: pilsicainide-effective, bepridil-effective, and failure (**Figure 1**). The FCL and FCL-CV data for the 3 groups were analyzed at baseline without any antiarrhythmic agents.

Protocol 2 (Comparison of the Change Caused by

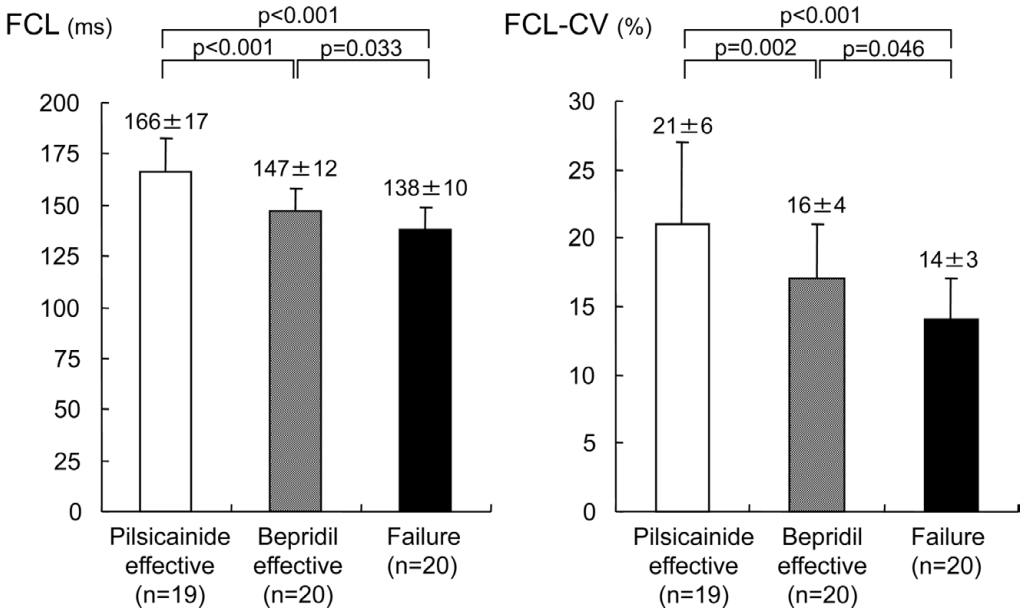

Figure 2. Comparison of the FCL and FCL-CV at baseline. The FCL was longest in the pilsicainide-effective group, moderate in the bepridil-effective group, and shortest in the failure group. Similarly, the FCL-CV was largest in the pilsicainide-effective group, moderate in the bepridil-effective group and smallest in the failure group. CV, coefficient of variation; FCL, fibrillation cycle length.

Drug Administration) In patients in whom AF terminated between the 2- and 4-week time points, an AF-ECG could be recorded only after the 2-week drug administration. In such patients, the prolongation of the FCL (ΔFCL = [FCL at 2 week] − [FCL at baseline]) and the increase in the FCL-CV (ΔFCL-CV) were calculated and compared with those from the failure group patients (**Figure 1**).

To obtain the most useful cut-off point for the prediction of drug efficacy in terminating AF, receiver-operating characteristic (ROC) curves were drawn for each parameter of the FCL. The sensitivity and specificity of each parameter were evaluated in the patients with pilsicainide or bepridil treatment.

Statistical Analysis

All values are expressed as the mean ± standard deviation. Statistical analysis was performed with 1-way ANOVA test. A P-value of <0.05 was considered significant.

Results

Pharmacological Cardioversion and Grouping

Of 59 patients, 28 were assigned to pilsicainide therapy and the remaining 31 were assigned to bepridil therapy in the first step (**Figure 1**). Subsequently, 12/28 patients undergoing pilsicainide therapy and 13/31 patients undergoing bepridil therapy achieved sinus rhythm and were classified into the pilsicainide- and bepridil-effective groups, respectively. In the remaining 34 patients in whom sinus rhythm could not be achieved in the first step, the antiarrhythmic therapies were crossed over and 16 were assigned to bepridil and 18 to pilsicainide. In the second step, the AF in 7/18 patients with pilsicainide therapy and 7/16 patients with bepridil therapy was successfully converted to sinus rhythm and these patients were also classified into the pilsicainide- and bepridil-effective groups, respectively. The remaining 20 patients in whom sinus rhythm could not be achieved with either drug were classified as the failure group. In all 20 patients, recovery of sinus rhythm was achieved by other therapies: combination of pilsicainide and bepridil (n=8) or direct current cardioversion (n=12).

In the first and second steps, pilsicainide and bepridil were administered to 46 and 47 patients, respectively. In the 46 patients with pilsicainide therapy, AF was interrupted during the initial 2 weeks in 7, in the following 2 weeks in 12, and was sustained in the remaining 27 patients. In the 47 patients with bepridil therapy, AF was interrupted during the initial 2 weeks in 4, in the following 2 weeks in 16, and was sustained in the remaining 27 patients (**Figure 1**). The ΔFCL and ΔFCL-CV were calculated in the patients in whom an AF-ECG at the 2-week time point could be recorded, and they were compared with those for the failure group patients.

Comparison of the FCL and FCL-CV Data

Figure 2 shows the FCL and FCL-CV data in the pilsicainide- and bepridil-effective groups and the failure group for AF termination. The FCL was longest in the pilsicainide-effective group, moderate in the bepridil-effective group and shortest in failure group, and a significant difference was observed among the 3 groups. Similarly, the FCL-CV was largest in the pilsicainide-effective group, moderate in the bepridil-effective group and smallest in the failure group, and the difference was also significant among the 3 groups.

Figure 3 is a comparison of the ΔFCL data of the pilsicainide-effective, bepridil-effective and the failure groups. In all patients, the FCL was prolonged by the 2-week antiarrhythmic drug therapy, even with pilsicainide or bepridil. Because the ΔFCL was calculated as the difference in the FCL from baseline to the 2-week time point, the change could be considered as FCL prolongation caused by the 2-week antiarrhythmic drug therapy in individual patients. The result was that the ΔFCL was longer in the pilsicainide-effective patients than in the failure patients (**Figure 3 Left**)

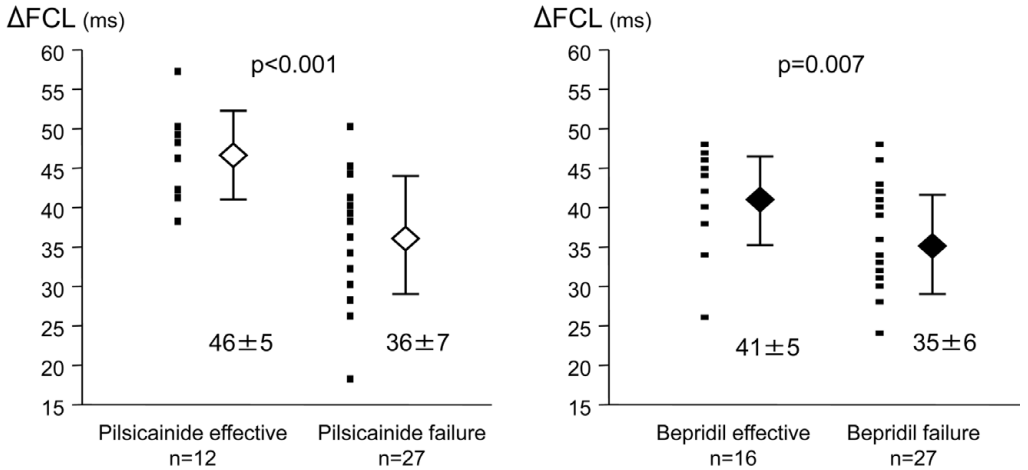

Figure 3. Comparison of the changes in the FCL after 2-week antiarrhythmic drug therapy. The ΔFCL indicates prolongation of the FCL after 2-week antiarrhythmic drug therapy in the individual patients, and it was larger in the pilsicainide-effective patients than in the failure patients (**Left**). The ΔFCL was also larger in the bepridil-effective patients than in the failure patients (**Right**). FCL, fibrillation cycle length.

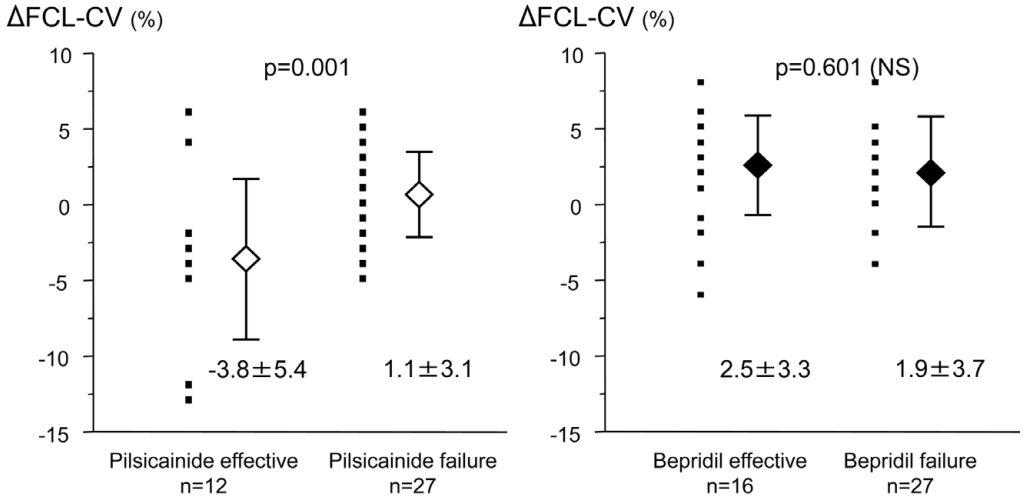

Figure 4. Comparison of the changes in the FCL-CV after 2-week antiarrhythmic drug therapy. The ΔFCL-CV indicates the change in the FCL-CV after 2-week antiarrhythmic drug therapy in the individual patients. The pilsicainide-effective patients had a larger negative number than the failure patients (**Left**). The ΔFCL-CV did not show any significant difference between the bepridil-effective and failure patients (**Right**). CV, coefficient of variation; FCL, fibrillation cycle length.

and was longer in the bepridil-effective patients than in the failure patients (**Figure 3 Right**).

Similarly, **Figure 4** is a comparison of the ΔFCL-CV data for the effective and failure patients with pilsicainide or bepridil. Because the FCL-CV is a parameter of the variability of the FCL, a decrease or increase in the ΔFCL-CV could be considered to indicate that of FCL variability caused by the 2-week antiarrhythmic drug therapy. The mean ΔFCL-CV data for the pilsicainide effective patients was negative and was larger than that in the failure patients (**Figure 4 Left**). In contrast, no significant difference in the ΔFCL-CV was observed between the bepridil effective and failure patients (**Figure 4 Right**).

Prediction of Drug Efficacy Using FCL and FCL-CV

Protocol 1 (FCL Data at Baseline) **Figure 5** shows the ROC curves for predicting AF termination using the FCL and FCL-CV data at baseline (ie, before drug administration). In the patients with pilsicainide therapy, statistically significant ROC curves were drawn from the FCL and FCL-CV data for predicting AF termination (**Figure 5 Left**). When the cut-off point was set at 148 ms for the FCL, successful interruption of the AF by pilsicainide could be predicted in patients with a longer FCL (sensitivity 0.917, specificity 0.612; P=0.007). Similarly, successful AF termination by pilsicainide could be predicted in patients with a larger FCL-CV, with a sensitivity of 0.583 and specificity of 0.833 when the cut-off point was set at 18% (P=0.015). In contrast, ROC curves could not be drawn for all the patients receiving bepridil therapy because AF was not terminated by bepridil in some patients with a longer FCL. However, in the second step, pilsicainide could terminate AF in all such patients with a longer FCL (ie, failure to react to bepridil in the first step, but successful termination of AF by pilsicainide in the second step; **Figure 1**). When patients in whom AF was terminated by pilsicainide were excluded from the study population, ROC curves could be drawn for the FCL and FCL-CV (**Figure 5 Right**). As a

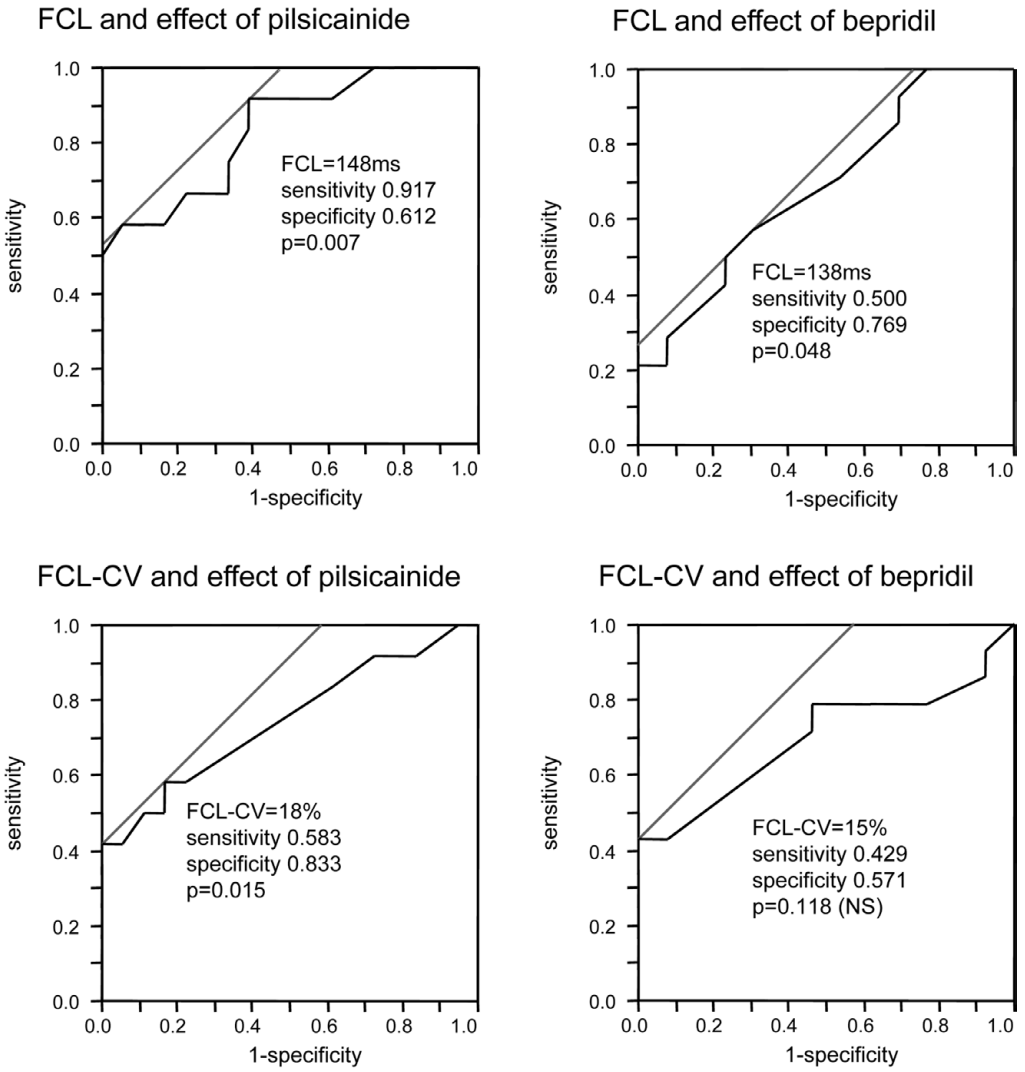

Figure 5. ROC curves for the prediction of AF termination using the FCL and FCL-CV data at baseline. ROC curves in all patients receiving (**Left**) pilsicainide therapy and (**Right**) bepridil therapy. AF termination with pilsicainide could be predicted in the patients with a FCL >148 ms or FCL-CV >18%. AF, atrial fibrillation; CV, coefficient of variation; FCL, fibrillation cycle length; ROC, receiver-operating characteristic.

result, AF termination by bepridil could be predicted in patients with a longer FCL with a sensitivity of 0.500 and specificity of 0.769 when the cut-off point was set at 138 ms (P=0.048). However, the ROC curve for the FCL-CV was not significant even in this selected population.

Protocol 2 (Change in the FCL Data During First 2 Weeks) **Figure 6** shows the ROC curves for predicting AF termination using the changes in the FCL and FCL-CV (ie, ΔFCL and ΔFCL-CV in the initial 2 weeks of drug administration). The ROC curves in the patients with pilsicainide therapy were statistically significant (**Figure 6 Left**). AF termination could be predicted in patients with a larger positive ΔFCL (ie, prolongation of the FCL), with a sensitivity of 0.875 and specificity of 0.833 when the cut-off point was set at 41 ms (P=0.001), or with a larger negative ΔFCL-CV (ie, decrease in the FCL-CV), with a sensitivity of 0.625 and specificity of 0.889 when the cut-off point was set at –4% (P=0.044). In the patients with bepridil therapy, the ROC curve for the ΔFCL was statistically significant, and AF termination could be predicted in patients with a larger ΔFCL, with a sensitivity of 0.846 and specificity of 0.655 when the cut-off point was set at 38 ms (P=0.034). In contrast, the ROC curve for the ΔFCL-CV was not significant in the patients with bepridil therapy.

Comparison of Bepridil-Failure/Pilsicainide-Effective Patients and Pilsicainide-Failure/Bepridil-Effective Patients

Additionally, the clinical and FCL parameters were compared between bepridil-failure/pilsicainide-effective patients (n=7) and pilsicainide-failure/bepridil-effective patients (n=7, **Figure 1**). Because it has been proven that 1 drug was effective but the other was ineffective in these 14 patients, we expected that this comparison might clarify the difference between the effects of pilsicainide and bepridil on AF. **Table 2** is a summary of the data comparison. The number of patients for the ΔFCL and ΔFCL-CV data were 5 and 6, respectively, because these data were obtained from patients who showed interruption of AF between the 2- and 4-week time points. As a result, FCL-CV at baseline was the only

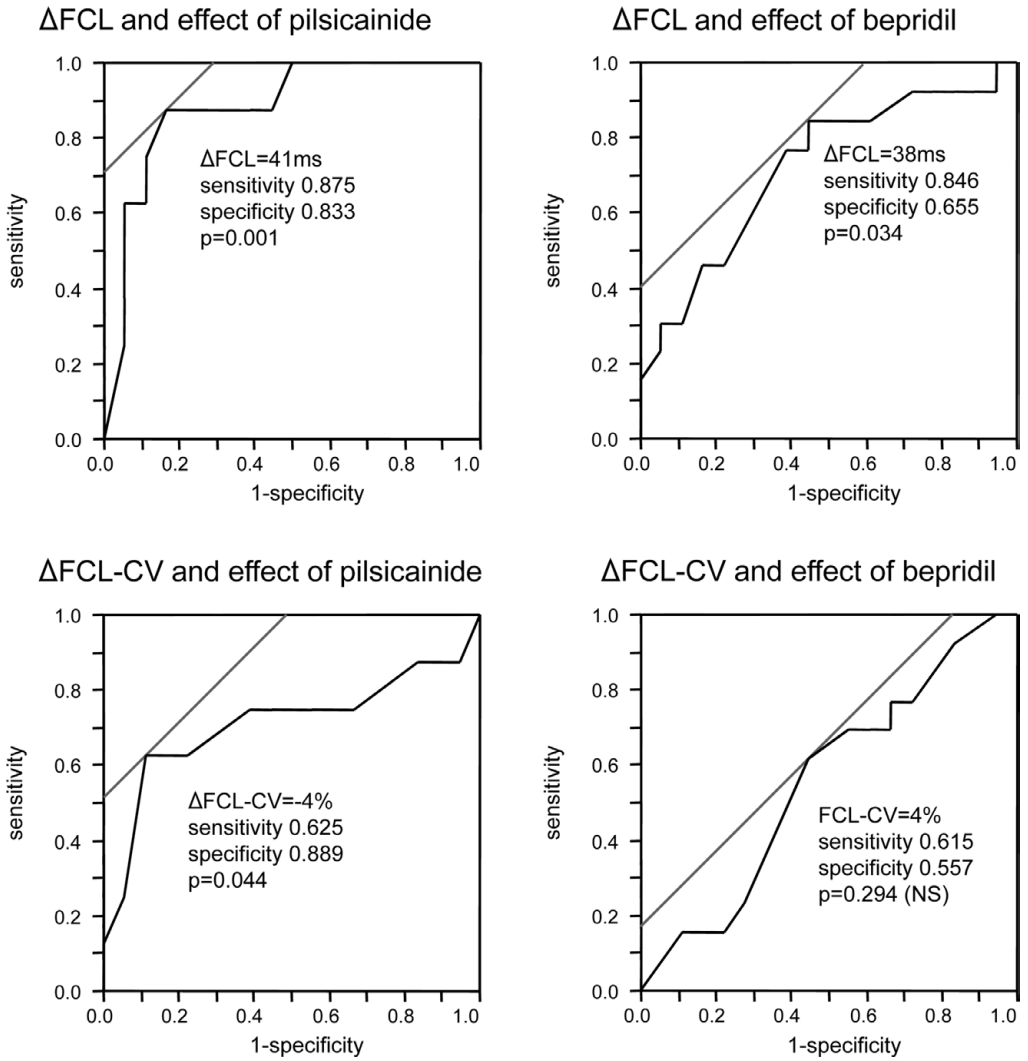

Figure 6. ROC curves for the prediction of AF termination using the changes in the FCL and FCL-CV during the initial 2 weeks of drug therapy. ROC curves in the patients with (**Left**) pilsicainide therapy and (**Right**) bepridil therapy. AF termination by pilsicainide could be predicted in the patients with a positive ΔFCL >41 ms or negative ΔFCL-CV >–4%. AF, atrial fibrillation; CV, coefficient of variation; FCL, fibrillation cycle length; ROC, receiver-operating characteristic.

Table 2. Comparison of Bepridil-Failure/Pilsicainide-Effective Patients and Pilsicainide-Failure/Bepridil-Effective Patients

	Bepridil-failure/ Pilsicainide-effective	Pilsicainide-failure/ Bepridil-effective	P value
n	7	7	
Age (years)	67±5	65±6	0.563
Gender (F/M)	3/4	2/5	0.577
LVEF (%)	63±6	61±5	0.551
LAd (mm)	40±3	41±4	0.588
FCL (ms)	169±20	149±13	0.055
FCL-CV (%)	24±6	18±3	0.029*
n	5	6	
ΔFCL (ms)	45±7	42±3	0.312
ΔFCL-CV (%)	–5±8	1±4	0.091

*Statistical significance.
Abbreviations see in Table 1.

parameter that exhibited significant difference between the 2 groups. FCL was larger in the bepridil-failure/pilsicainide-effective patients than in the pilsicainide-failure/bepridil-effective patients, but the difference was not significant, possibly because of the small patient population.

Discussion

Clinical Importance of Predicting the Antiarrhythmic Drug Effect in AF

Although there is a theoretical background for the termi-

nation and prevention of AF by antiarrhythmic agents,[10,13,15] the clinical effects in individual patients can be determined only by trial and error, because the electrophysiological properties of the AF substrate vary in each patient. However, this method of terminating AF will increase the chance of a cardiogenic embolism and the AF itself can become more difficult to terminate because of the progression of electrical remodeling.[16,17] Therefore, it is useful to predict the efficacy of an antiarrhythmic agent from the beginning of a therapeutic trial. It is empirically known that AF with coarse fibrillation waves can be more easily terminated with sodium-channel blockers,[9,10] probably because the reentrant circuits of this type of AF are simpler than those of AF with fine fibrillation waves, and a single conduction block may more easily result in the complete termination of the AF. However, to the best of our knowledge, there is not a feasible or objective index for predicting the efficacy of antiarrhythmic agents from the fibrillation waves occurring at baseline.

In the present study, spectral analysis of the fibrillation waves in the surface ECG was performed and the parameters of the fibrillation waves (ie, FCL and FCL-CV) differed among the pilsicainide- and bepridil-effective groups and failure patients. This finding indicates that FCL and FCL-CV, which were calculated in the analysis of the fibrillation wave, reflect the atrial electrophysiological properties and might be useful for distinguishing patients who will have a different response to a specific antiarrhythmic agent. Spatial or time-dependent variation in FCL data might also affect the result, but because we used mean data of FCL from 20 consecutive analysis windows, the influence of transient changes in FCL data is considered to be small.[6–8] In addition, because only patients with PAF were included in this study, time-dependent variations in FCL and FCL-CV data were also considered small.[6]

Prediction of Drug Efficacy by Fibrillation Wave Analysis

In the present study, we were able to set a good cut-off point for predicting the efficacy of pilsicainide, a pure sodium-channel blocker, for the termination of PAF by using the indices of FCL and FCL-CV at baseline before antiarrhythmic therapy. The changes in the FCL and FCL-CV data caused by 2 weeks' pilsicainide therapy (ie, prolongation of the FCL and decrease in the FCL-CV) were also useful as indices for predicing the efficacy of the antiarrhythmic therapy during further continuation of the drug administration. The latter assumption seems to be reasonable because the changes in those parameters are considered to reflect the action of pilsicainide on the atrial electrophysiological properties,[8] and the parameters might be useful for deciding whether to continue or discontinue pilsicainide as therapy to terminate AF. In contrast, the former seems to be more interesting and useful because drug efficacy could be predicted before the practical trial of the antiarrhythmic agent.

Because the fibrillation waves in the surface ECG may reflect multiple wave fronts existing simultaneously on the atrial surface, we cannot conclude that the FCL in our measurements directly reflected the actual CL of local atrial activation.[7,8,14] However, previous reports have documented a good correlation between FCL with our measurement technique and the actual atrial activation,[7,8] and therefore the FCL used in this study was considered to represent the average of the total atrial activation frequency. In contrast, the FCL-CV reflects the variability of the FCL during the time course of the recording,[8] and it may reflect the instability of the individual reentrant circuits (ie, disappearance and construction of larger reentrant circuits of AF). Therefore, a larger FCL and greater FCL-CV may be objective indices of "coarse fibrillation waves" on the surface ECG, and be good markers of the efficacy of pilsicainide in terminating AF. Additionally, the effectiveness of pilsicainide for AF prevention after interruption of AF should also be considered to understand the result of this study. Because pilsicainide may block the exit of premature atrial contractions originating from the pulmonary vein, it would reduce the trigger of AF and thus result in maintaining sinus rhythm. However, this effect of pilsicainide cannot be predicted by analysis of the fibrillation waves.

In contrast to pilsicainide, the bepridil trial gave more complicated results. Statistically significant cut-off points could be set for the FCL and ΔFCL data after excluding the pilsicainide-effective patients, but not for the FCL-CV and ΔFCL-CV data. Further, even for the FCL and ΔFCL data, the sensitivities and specificities of those parameters were lower than those for pilsicainide therapy. These results indicate that the FCL parameters could be useful for distinguishing bepridil-effective patients and failure patients, but the efficacy of using the indices would be smaller than for pilsicainide therapy.

Bepridil is a multi-channel blocker with a relatively strong effect on the I_{kr}, I_{ks}, I_{to}, and L-Ca channels and a weak effect on the I_{Na} and T-Ca channels.[11,12,18] Its antiarrhythmic effect has been recently emphasized, especially for long-lasting PAF.[11] The main action of bepridil as an antiarrhythmic agent is considered to be a potassium-channel blocking effect (ie, prolongation of the action potential).[13] Because some investigators have suggested that bepridil would need a longer period of therapy (up to 2–3 months) to show its real antiarrhythmic effect,[18–20] the 4-week protocol in this study might have underestimated the drug's efficacy for AF termination. However, the effect of bepridil for terminating AF in this study was well-matched with pilsicainide, and bepridil was effective in patients with a shorter FCL and smaller FCL-CV than was pilsicainide. It seems to be difficult to extract a feasible index for the efficacy of bepridil from this data, but the FCL and FCL-CV data might be useful for screening out bepridil-resistant patients (ie, refractory to class III agents). Additionally, even though the effect was smaller than that for pilsicainide, the prolongation of the FCL after the 2-week bepridil therapy was a useful index for predicting further efficacy, so the decision to continue or discontinue therapy could be made more objectively. Besides the effect of bepridil on AF interruption, its effect on AF prevention should also be considered. Because bepridil prolongs atrial refractoriness, it would prevent the formation of AF reentrant circuits and prevent recurrence of the arrhythmia after interruption of AF. Because the parameters of FCL and FCL-CV do not directly reflect atrial refractoriness, especially during the sinus rhythm, in this study the preventive effect of bepridil for AF could not be predicted by FCL data.

Action of Antiarrhythmic Agents on FCL

Class I antiarrhythmic agents prolong the CL of reentrant rhythms by decreasing the conduction velocity, especially for anatomically determined reentrant circuits.[21] In contrast, class III antiarrhythmic agents do not affect conduction velocity, but enlarge the central line of block as a result of prolongation of the action potential duration and lead to

prolongation of the CL, especially in spiral or leading circuit reentry.[22,23] Because the arrhythmogenic substrate of AF is considered to be complex anatomically and functionally determine the reentrant circuits,[3,21] both class I and III antiarrhythmic agents may prolong the CL of the reentrant rhythm, and the response to the antiarrhythmic agents is considered to reflect the electrophysiological properties of the dominant reentrant circuit. Because the FCL in this study was an indirect but averaged index of atrial activation frequency, the change in the FCL caused by the antiarrhythmic agent might well reflect the action of the drug on the dominant reentrant circuit and may have led to the good results for the FCL data in this study as an index for predicting drug efficacy for AF termination. In contrast, the FCL-CV data is more complicated. Because this parameter was calculated from the standard deviation of FCLs of 20 continuous epochs,[8] it is considered to reflect the stability of the reentrant circuits over time. Because a larger FCL-CV indicates a larger variation in FCL, an AF patient with a larger FCL-CV may have more chance of momentary larger FCLs for which pilsicainide is effective for AF interruption. Therefore, a larger FCL-CV can be an index for predicting the effective result of pilsicainide for AF interruption. Interestingly, FCL-CV was the only parameter that exhibited a significant difference between bepridil-failure/pilsicainide-effective patients and pilsicainide-failure/bepridil-effective patients (**Table 2**). That result also indicates the importance of FCL-CV as a predictor of AF interruption by antiarrhythmic agents.

Recently, it has been reported that bepridil might have an effect on atrial electrical remodeling,[24–26] similar to amiodarone.[27] In the present study, the effect of pilsicainide and bepridil on electrical remodeling was not evaluated, but it seems that, practically, it would have been difficult to monitor the atrial refractory period without repeating the cardiac catheterization. The evaluation of the atrial electrophysiological properties using FCL measurement might become a possible method for monitoring atrial electrical remodeling,[28] but such an evaluation should be performed for a longer period than 4 weeks.

Study Limitations

First, the number of patients was limited, thus though drug-effective patients in the 1st and 2nd steps may differ from each other, the FCL and FCL-CV data should be analyzed separately; however, this could not be done because of the small number of patients. Second, the antiarrhythmic agent was crossed over in the failure patients in the first step, and we also used similar data in the second step. Although a 2-week washout period was set between the first and second steps, the influence of the first drug could not be fully ruled out, especially during the baseline evaluation in the second step. Third, because we only evaluated the efficacy of 2 selected antiarrhythmic agents for PAF, other popular antiarrhythmic agents, such as amiodarone, flecainide, propafenone etc, should be evaluated if possible. Finally, because the entire observation period was 10 weeks in total in some patients, the influence of the progression of electrical remodeling during that period cannot be ruled out. These points should be solved in future studies with a different design.

Conclusions

In a spectral analysis of the fibrillation waves in PAF, the FCL and FCL-CV data differed among patients with different results for drug efficacy in terminating AF. AF interruption by pilsicainide or bepridil could be predicted from the baseline data or from the changes after a 2-week drug therapy regimen. The electrophysiological data of the fibrillation waves might be used as feasible indices for determining the most effective antiarrhythmic therapy for PAF.

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Disclosure

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